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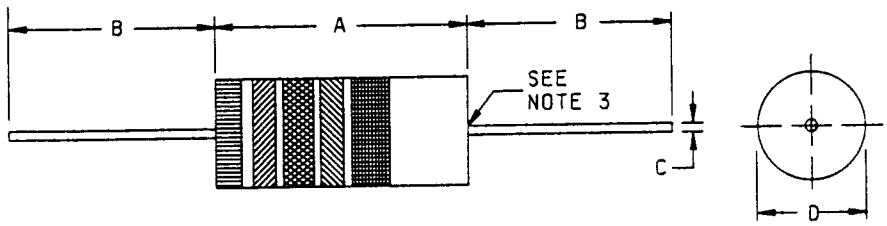
MIL-PRF-22684/8B
31 January 1978
SUPERSEDING
MIL-R-22684/8A(USAF)
7 September 1967

PERFORMANCE SPECIFICATION SHEET

RESISTORS, FIXED, FILM, INSULATED, STYLE RL42. . . .TX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of Specification MIL-R-22684.



	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.648	.728	16.46	18.49
B	1.375	1.625	34.92	41.28
C	.043	.047	1.09	1.19
D	.300	.336	7.62	8.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

FSC 5905

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Table I. Part Number Designation

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL42S100GTX	10	G	S	1001	RL42S331GTX	330	G	S	1073
RL42S100JTX	10	J	S	1002	RL42S331JTX	330	J	S	1074
RL42S110GTX	11	G	S	1003	RL42S361GTX	360	G	S	1075
RL42S110JTX	11	J	S	1004	RL42S361JTX	360	J	S	1076
RL42S120GTX	12	G	S	1005	RL42S391GTX	390	G	S	1077
RL42S120JTX	12	J	S	1006	RL42S391JTX	390	J	S	1078
RL42S130GTX	13	G	S	1007	RL42S431GTX	430	G	S	1079
RL42S130JTX	13	J	S	1008	RL42S431JTX	430	J	S	1080
RL42S150GTX	15	G	S	1009	RL42S471GTX	470	G	S	1081
RL42S150JTX	15	J	S	1010	RL42S471JTX	470	J	S	1082
RL42S160GTX	16	G	S	1011	RL42S511GTX	510	G	S	1083
RL42S160JTX	16	J	S	1012	RL42S511JTX	510	J	S	1084
RL42S180GTX	18	G	S	1013	RL42S561GTX	560	G	S	1085
RL42S180JTX	18	J	S	1014	RL42S561JTX	560	J	S	1086
RL42S200GTX	20	G	S	1015	RL42S621GTX	620	G	S	1087
RL42S200JTX	20	J	S	1016	RL42S621JTX	620	J	S	1088
RL42S220GTX	22	G	S	1017	RL42S681GTX	680	G	S	1089
RL42S220JTX	22	J	S	1018	RL42S681JTX	680	J	S	1090
RL42S240GTX	24	G	S	1019	RL42S751GTX	750	G	S	1091
RL42S240JTX	24	J	S	1020	RL42S751JTX	750	J	S	1092
RL42S270GTX	27	G	S	1021	RL42S821GTX	820	G	S	1093
RL42S270JTX	27	J	S	1022	RL42S821JTX	820	J	S	1094
RL42S300GTX	30	G	S	1023	RL42S911GTX	910	G	S	1095
RL42S300JTX	30	J	S	1024	RL42S911JTX	910	J	S	1096
RL42S330GTX	33	G	S	1025	RL42S102GTX	1,000	G	S	1097
RL42S330JTX	33	J	S	1026	RL42S102JTX	1,000	J	S	1098
RL42S360GTX	36	G	S	1027	RL42S112GTX	1,100	G	S	1099
RL42S360JTX	36	J	S	1028	RL42S112JTX	1,100	J	S	1100
RL42S390GTX	39	G	S	1029	RL42S122GTX	1,200	G	S	1101
RL42S390JTX	39	J	S	1030	RL42S122JTX	1,200	J	S	1102
RL42S430GTX	43	G	S	1031	RL42S132GTX	1,300	G	S	1103
RL42S430JTX	43	J	S	1032	RL42S132JTX	1,300	J	S	1104
RL42S470GTX	47	G	S	1033	RL42S152GTX	1,500	G	S	1105
RL42S470JTX	47	J	S	1034	RL42S152JTX	1,500	J	S	1106
RL42S510GTX	51	G	S	1035	RL42S162GTX	1,600	G	S	1107
RL42S510JTX	51	J	S	1036	RL42S162JTX	1,600	J	S	1108
RL42S560GTX	56	G	S	1037	RL42S182GTX	1,800	G	S	1109
RL42S560JTX	56	J	S	1038	RL42S182JTX	1,800	J	S	1110
RL42S620GTX	62	G	S	1039	RL42S202GTX	2,000	G	S	1111
RL42S620JTX	62	J	S	1040	RL42S202JTX	2,000	J	S	1112
RL42S680GTX	68	G	S	1041	RL42S222GTX	2,200	G	S	1113
RL42S680JTX	68	J	S	1042	RL42S222JTX	2,200	J	S	1114
RL42S750GTX	75	G	S	1043	RL42S242GTX	2,400	G	S	1115
RL42S750JTX	75	J	S	1044	RL42S242JTX	2,400	J	S	1116
RL42S820GTX	82	G	S	1045	RL42S272GTX	2,700	G	S	1117
RL42S820JTX	82	J	S	1046	RL42S272JTX	2,700	J	S	1118
RL42S910GTX	91	G	S	1047	RL42S302GTX	3,000	G	S	1119
RL42S910JTX	91	J	S	1048	RL42S302JTX	3,000	J	S	1120
RL42S101GTX	100	G	S	1049	RL42S332GTX	3,300	G	S	1121
RL42S101JTX	100	J	S	1050	RL42S332JTX	3,300	J	S	1122
RL42S111GTX	110	G	S	1051	RL42S362GTX	3,600	G	S	1123
RL42S111JTX	110	J	S	1052	RL42S362JTX	3,600	J	S	1124
RL42S121GTX	120	G	S	1053	RL42S392GTX	3,900	G	S	1125
RL42S121JTX	120	J	S	1054	RL42S392JTX	3,900	J	S	1126
RL42S131GTX	130	G	S	1055	RL42S432GTX	4,300	G	S	1127
RL42S131JTX	130	J	S	1056	RL42S432JTX	4,300	J	S	1128
RL42S151GTX	150	G	S	1057	RL42S472GTX	4,700	G	S	1129
RL42S151JTX	150	J	S	1058	RL42S472JTX	4,700	J	S	1130
RL42S161GTX	160	G	S	1059	RL42S512GTX	5,100	G	S	1131
RL42S161JTX	160	J	S	1060	RL42S512JTX	5,100	J	S	1132
RL42S181GTX	180	G	S	1061	RL42S562GTX	5,600	G	S	1133
RL42S181JTX	180	J	S	1062	RL42S562JTX	5,600	J	S	1134
RL42S201GTX	200	G	S	1063	RL42S622GTX	6,200	G	S	1135
RL42S201JTX	200	J	S	1064	RL42S622JTX	6,200	J	S	1136
RL42S221GTX	220	G	S	1065	RL42S682GTX	6,800	G	S	1137
RL42S221JTX	220	J	S	1066	RL42S682JTX	6,800	J	S	1138
RL42S241GTX	240	G	S	1067	RL42S752GTX	7,500	G	S	1139
RL42S241JTX	240	J	S	1068	RL42S752JTX	7,500	J	S	1140
RL42S271GTX	270	G	S	1069	RL42S822GTX	8,200	G	S	1141
RL42S271JTX	270	J	S	1070	RL42S822JTX	8,200	J	S	1142
RL42S301GTX	300	G	S	1071	RL42S912GTX	9,100	G	S	1143
RL42S301JTX	300	J	S	1072	RL42S912JTX	9,100	J	S	1144

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Table I. Part Number Designation (Cont'd)

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Megohm			
RL42S103GTX	10,000	G	S	1145	RL42S124GTX	0.12	G	S	1197
RL42S103JTX	10,000	J	S	1146	RL42S124JTX	0.12	J	S	1198
RL42S113GTX	11,000	G	S	1147	RL42S134GTX	0.13	G	S	1199
RL42S113JTX	11,000	J	S	1148	RL42S134JTX	0.13	J	S	1200
RL42S123GTX	12,000	G	S	1149	RL42S154GTX	0.15	G	S	1201
RL42S123JTX	12,000	J	S	1150	RL42S154JTX	0.15	J	S	1202
RL42S133GTX	13,000	G	S	1151	RL42S164GTX	0.16	G	S	1203
RL42S133JTX	13,000	J	S	1152	RL42S164JTX	0.16	J	S	1204
RL42S153GTX	15,000	G	S	1153	RL42S184GTX	0.18	G	S	1205
RL42S153JTX	15,000	J	S	1154	RL42S184JTX	0.18	J	S	1206
RL42S163GTX	16,000	G	S	1155	RL42S204GTX	0.20	G	S	1207
RL42S163JTX	16,000	J	S	1156	RL42S204JTX	0.20	J	S	1208
RL42S183GTX	18,000	G	S	1157	RL42S224GTX	0.22	G	S	1209
RL42S183JTX	18,000	J	S	1158	RL42S224JTX	0.22	J	S	1210
RL42S203GTX	20,000	G	S	1159	RL42S244GTX	0.24	G	S	1211
RL42S203JTX	20,000	J	S	1160	RL42S244JTX	0.24	J	S	1212
RL42S223GTX	22,000	G	S	1161	RL42S274GTX	0.27	G	S	1213
RL42S223JTX	22,000	J	S	1162	RL42S274JTX	0.27	J	S	1214
RL42S243GTX	24,000	G	S	1163	RL42S304GTX	0.30	G	S	1215
RL42S243JTX	24,000	J	S	1164	RL42S304JTX	0.30	J	S	1216
RL42S273GTX	27,000	G	S	1165	RL42S334GTX	0.33	G	S	1217
RL42S273JTX	27,000	J	S	1166	RL42S334JTX	0.33	J	S	1218
RL42S303GTX	30,000	G	S	1167	RL42S364GTX	0.36	G	S	1219
RL42S303JTX	30,000	J	S	1168	RL42S364JTX	0.36	J	S	1220
RL42S333GTX	33,000	G	S	1169	RL42S394GTX	0.39	G	S	1221
RL42S333JTX	33,000	J	S	1170	RL42S394JTX	0.39	J	S	1222
RL42S363GTX	36,000	G	S	1171	RL42S434GTX	0.43	G	S	1223
RL42S363JTX	36,000	J	S	1172	RL42S434JTX	0.43	J	S	1224
RL42S393GTX	39,000	G	S	1173	RL42S474GTX	0.47	G	S	1225
RL42S393JTX	39,000	J	S	1174	RL42S474JTX	0.47	J	S	1226
RL42S433GTX	43,000	G	S	1175	RL42S514GTX	0.51	G	S	1227
RL42S433JTX	43,000	J	S	1176	RL42S514JTX	0.51	J	S	1228
RL42S473GTX	47,000	G	S	1177	RL42S564GTX	0.56	G	S	1229
RL42S473JTX	47,000	J	S	1178	RL42S564JTX	0.56	J	S	1230
RL42S513GTX	51,000	G	S	1179	RL42S624GTX	0.62	G	S	1231
RL42S513JTX	51,000	J	S	1180	RL42S624JTX	0.62	J	S	1232
RL42S563GTX	56,000	G	S	1181	RL42S684GTX	0.68	G	S	1233
RL42S563JTX	56,000	J	S	1182	RL42S684JTX	0.68	J	S	1234
RL42S623GTX	62,000	G	S	1183	RL42S754GTX	0.75	G	S	1235
RL42S623JTX	62,000	J	S	1184	RL42S754JTX	0.75	J	S	1236
RL42S683GTX	68,000	G	S	1185	RL42S824GTX	0.82	G	S	1237
RL42S683JTX	68,000	J	S	1186	RL42S824JTX	0.82	J	S	1238
RL42S753GTX	75,000	G	S	1187	RL42S914GTX	0.91	G	S	1239
RL42S753JTX	75,000	J	S	1188	RL42S914JTX	0.91	J	S	1240
RL42S823GTX	82,000	G	S	1189	RL42S105GTX	1.0	G	S	1241
RL42S823JTX	82,000	J	S	1190	RL42S105JTX	1.0	J	S	1242
RL42S913GTX	91,000	G	S	1191	RL42S115GTX	1.1	G	S	1243
RL42S913JTX	91,000	J	S	1192	RL42S115JTX	1.1	J	S	1244
	Megohm				RL42S125GTX	1.2	G	S	1245
RL42S104GTX	0.10	G	S	1193	RL42S125JTX	1.2	J	S	1246
RL42S104JTX	0.10	J	S	1194	RL42S135GTX	1.3	G	S	1247
RL42S114GTX	0.11	G	S	1195	RL42S135JTX	1.3	J	S	1248
RL42S114JTX	0.11	J	S	1196	RL42S155GTX	1.5	G	S	1249
					RL42S155JTX	1.5	J	S	1250

APPLICABLE DOCUMENT

MIL-STD-1276 Weldable Leads for Electronic Parts

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Resistance values: See table I.
Resistance tolerances: See table I.
Terminal type: See table I.
Minimum resistance value: 10 ohms.
Maximum resistance value: 1.5 megohm.

Maximum continuous working voltage: 500 volts.

Power rating: 2 watts at 70°C.

Thermal shock: Method 107 of MIL-STD-202, condition A, except that 150° +3°, -0°C shall be used instead of 85° +3°, -0°C.
Change in resistance shall not exceed ±(1 percent + 0.05 ohm).

Low temperature operation:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Short time overload:
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Terminal strength: Method 211 of MIL-STD-202.
Condition A-Pull test 5 pounds.
Condition D-Twist test: rotated 360° for 3 rotations.
Change in resistance shall not exceed ±(0.5 percent + 0.05 ohm).

Dielectric withstanding voltage: Methods 301 and 105 of MIL-STD-202.
Atmospheric pressure-1,000 volts rms.
Barometric pressure-500 volts rms.

Insulation resistance: Method 302 of MIL-STD-202.
Condition A or B, whichever is practicable-Insulation resistance shall be not less than 100 megohms.

Resistance to soldering heat: Method 210 of MIL-STD-202.
Condition A-Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Moisture resistance: Method 106 of MIL-STD-202.
Change in resistance shall not exceed 2.0 percent.
Insulation resistance shall be 100 megohms, minimum.

Resistance temperature characteristic: Method 304 of MIL-STD-202.
Change in resistance shall not exceed ±0.020 percent per degree C(200 ppm/°C).

Life: Method 108 of MIL-STD-202.
Condition D-Change in resistance shall not exceed 3.0 percent.

Shock (specified pulse) - Method 213 of MIL-STD-202.
Condition I-Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Vibration, high frequency: Method 204 of MIL-STD-202.
Condition D-Change in resistance shall not exceed ±(0.5 percent +0.05 ohm).

Solderability: Method 208 of MIL-STD-202.

Lead material: Type C, in accordance with MIL-STD-1276.

Lead coating: Tin-lead (Sn 30 to 70 percent) 0.0005 to 0.0015 inch thickness. The maximum thickness shall not exceed twice the minimum thickness at any cross section. The leads shall be free of nonconductive and foreign material beyond the maximum body dimensions.

QUALITY ASSURANCE PROVISIONS

Sampling and inspection: MIL-R-22684 and as specified herein.

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Qualification inspection: MIL-R-22684 on the respective non-TX version. (The TX version shall be procured from sources listed in QPL-22684 for the respective non-TX version.)

Quality conformance inspection: Inspection of product for delivery: Groups A, B, and C inspections. Delivery of shipment shall not be delayed pending completion of group C testing.

Group A inspection: MIL-R-22684, except power conditioning shall be performed on 100 percent of the product supplied under this specification.

Group B inspection: MIL-R-22684, except that the AQL shall be 2.5 (percent defective).

Group C inspection: MIL-R-22684. Successful performance of group C inspection for the respective non-TX version will suffice to demonstrate conformance to this requirement for TX version.

Power conditioning

Measurement before conditioning: DC resistance measured in accordance with paragraph entitled "DC Resistance" of MIL-R-22684.

Conditioning procedures: Each resistor supplied under this specification shall be subjected to power conditioning for a period of 24 hours at 150 percent rated power. Mounting shall be such that 25° ±2°C ambient air can circulate around the resistors by means of normal convection. The maximum applied voltage shall not exceed 500 V rms.

Measurement after conditioning: After power conditioning, the resistors shall be stabilized at room conditions for a minimum of 2 hours before the dc resistance measurements are made. Resistors that are outside of the ±2 percent tolerance limit and those that have changed more than 0.5 percent due to the conditioning shall be removed from the lot. When the combined quantity removed exceeds 5 percent of the lot, the entire lot shall be rejected. The supplier shall have a complete record available of all test data obtained during power conditioning testing (see ordering data).

TX identification: Resistors procured to, and meeting all of the criteria specified herein shall bear a green colored band in place of the band "E" in accordance with appendix B to MIL-R-22684. The symbol "TX" shall be printed at one of the locations specified below:

a. Preferred: On the green colored band described above.
b. Alternate: On the packing which packaged the items in groups.

In the event the resistor sample fails to meet the requirements of MIL-R-22684 and this specification sheet, the manufacturer shall remove the green band from the sample tested and also from all the resistors represented by the sample.

Qualification: The activity responsible for the Qualified Products List is the Engineering and Technical Support Division, ATTN: AFLC/LOIE, Wright Patterson Air Force Base, Ohio 45433; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444.

Ordering data: The procurement document should specify whether the power conditioning data is to be supplied.

Usage data: MIL-R-22684/8A(USAF) previously inactivated for new design as of 20 May 1969, reactivated for new design as of 18 November 1977.

Part number: Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M22684/08-1001

Basic Document Number	Dash number from Table I.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC, CG
Air Force - 19

Preparing activity:
Air Force - 85

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